DATA ARCHITECTURE BREAKDOWN

Databricks vs. Snowflake vs. Microsoft Fabric

Under-the-hood engine internals, storage formats, compute virtualization, and architectural trade-offs.

Databricks

THE CODE-FIRST LAKEHOUSE

1 Execution Engine: Photon (C++)

- Apache Spark + Vectorized Photon engine in C++
- JVM memory bypass via native SIMD instructions
- Extreme throughput Spark Structured Streaming

2 Storage: Delta Lake (UniForm)

- Parquet + JSON ACID transaction log (_delta_log)
- UniForm generates Iceberg & Hudi metadata in-place
- True open files in customer storage (S3/ADLS/GCS)

3 Governance: Unity Catalog

- 3-level namespace: catalog.schema.table
- Unified multi-cloud governance (AWS, Azure, GCP)
- Governs tables, models, features, functions & agents

4 AI & Workflows: Mosaic AI

- Native MLflow tracking + Vector Search + Model Serving
- Databricks Asset Bundles (DABs) for Git CI/CD
- Full LLM fine-tuning, RAG ops & Compound AI systems

ARCHITECTURAL SWEET SPOT & TRADEOFF

Best when: You write PySpark, Scala, or Python.

Unmatched for complex ETL pipelines, massive streaming data, model training, and cross-cloud open-lake architectures.

Trade-off: Cluster spin-up latency & DBU cost tuning.

Snowflake

THE SQL-FIRST WAREHOUSE

1 Compute: Virtual Warehouses

- Multi-cluster shared data (MCSD) architecture
- Zero cluster config: T-shirt sizing (XS to 6XL)
- Sub-second auto-resume & auto-suspend by query

2 Storage: Micro-Partitions + Iceberg

- Proprietary columnar micro-partitions (50-500MB)
- Automatic clustering & min/max pruning (zero index ops)
- Apache Iceberg tables backed by customer external storage

3 Governance: Snowflake Horizon

- Zero-copy cloning for instant dev/QA testing
- Secure Data Sharing across clouds with zero ETL copying
- Dynamic data masking, tag policies & row-level security

4 AI & Apps: Snowpark & Cortex

- Cortex LLMs callable directly inside SQL statements
- Snowpark pushes Python DataFrames into warehouse nodes
- Native App Framework + embedded Streamlit UIs

ARCHITECTURAL SWEET SPOT & TRADEOFF

Best when: You want pure SQL & zero infra ops.

The gold standard for analytical SQL, fast dbt execution, unlimited read concurrency, and data monetization.

Trade-off: Fast credit consumption if warehouse auto-suspend lags.

Microsoft Fabric

THE ALL-IN-ONE ECOSYSTEM

1 Compute: Shared Multi-Engine

- F-SKU capacity pool shared across all engines
- Apache Spark, T-SQL Warehouse, KQL & Data Factory
- 24-hour background smoothing absorbs bursty workloads

2 Storage: OneLake (SaaS Delta)

- Single logical SaaS data lake built on Delta Parquet
- OneLake Shortcuts virtualize ADLS, S3, & Snowflake
- Direct Lake mode: Power BI queries Delta with zero refresh

3 Governance: OneLake & Purview

- Domain & Workspace Data Mesh boundary isolation
- OneLake Security: Unified RBAC across all query engines
- Native automated Microsoft Purview catalog & lineage

4 AI & BI Integration: Copilot

- Built-in Copilot across Notebooks, SQL, and Power BI
- Native Git Integration (ADO & GitHub) with PBIR / NB sync
- Unrivaled seamless delivery to Microsoft 365 & Teams

ARCHITECTURAL SWEET SPOT & TRADEOFF

Best when: Your enterprise runs on Power BI & Azure.

The all-in-one SaaS platform: pipelines, Spark, warehouse, and Power BI under one tenant with Direct Lake speed.

Trade-off: Fixed capacity sizing & ecosystem still maturing.

THE KEY TAKEAWAY

The Real Architectural Trade-Off: Engine Control vs. Concurrency Simplicity vs. All-in-One SaaS

Databricks

Maximum compute control & deep open-source tuning.

Ideal for large engineering teams writing PySpark & training ML models.

Snowflake

Zero infra operations & unmatched SQL concurrency.

Ideal for analytics warehouses, fast dbt models, and cross-cloud sharing.

Microsoft Fabric

Unified SaaS ecosystem with zero-copy OneLake shortcuts.

Ideal for enterprises building reports in Power BI and living inside Azure.

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